

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Justification

Annotator-population mismatch is the highest-confidence concern in this analysis. Sentiment analysis annotation achieved Cohen's Kappa of only 0.45 [Q27] — below the 0.60–0.80 range typical for robust NLU annotation — with disagreements resolved by annotators agreeing among themselves [Q28]. Annotators beyond SA are described as 'student assistants' [Q114]; no evidence of professional journalist or editor involvement, despite the deployment explicitly requiring journalistic-register ground truth [A3]. RTE labels are AI-assisted (ChatGPT-5.1 improvement [Q61], ChatGPT-5-MINI quality and label verification [Q63, Q64]), with manual correction only where the LLM altered semantics [Q65]. JUDGEWEL NER uses LLMs as judges with minimal human verification [Q37]. The benchmark itself warns 'models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices' [Q120]. Given OC is a HIGH-priority dimension and the deployment requires journalist-register ground truth, score is 2.

Strengths

- The Lothritz et al. (2022) RTL NER subcorpus is fully human-annotated and high-precision [Q40, Q42]
- RTE labels underwent two-stage AI verification with manual correction where semantics drifted [Q62, Q65], offering some quality control
- The benchmark explicitly cautions deployers against treating performance as cultural/demographic coverage [Q121], enabling informed risk assessment

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels do not reflect professional Luxembourgish journalist/editor perspectives. The deployment requires the flexible journalistic register [A3], but annotators are described as student assistants [Q114] with no documented journalism background; LA labels derive from formal LOD/Spellchecker sources [Q30, Q31] and RTE labels are AI-assisted [Q63, Q64].

ID	Evidence
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies

OC-2: Likely substantial disagreement between benchmark annotators and the target population (working journalists/editors), particularly on borderline LA Ortho cases [A3] where the deployment explicitly anticipates formal-vs-journalistic disagreement.

ID	Evidence
----	----------

WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies
--------	--

OC-3: Annotator demographics: SA was annotated by two native Luxembourgish speakers [Q25] with Cohen's Kappa 0.45 [Q27]; intent translation by one native speaker with consultation [Q52]; other tasks by 'student assistants' [Q114]. No professional-journalist annotators documented; no Datasheet-level demographic breakdown provided.

OC-4: Re-annotation of LA, HA, and a sample of NER by professional Luxembourgish editors applying the deployment's flexible register would be necessary to establish convergent validity for the deployment.

OC-5: SA disagreements resolved by annotators agreeing among themselves [Q28] — a consensus aggregation that erases minority annotator perspectives without documented arbitration. Cohen's Kappa 0.45 [Q27] indicates substantial baseline disagreement subsequently flattened.

OC-6: Documented label issues likely to harm convergent and external validity: (a) student-annotator pool [Q114] vs. professional-journalist target users [A3]; (b) AI-assisted RTE labels [Q63, Q64]; (c) LLM-as-judge JUDGEWEL [Q37]; (d) low SA IAA [Q27]; (e) explicit benchmark caution against demographic-coverage claims [Q120, Q121].

Information Gaps

- Number, demographics, and journalism background of student annotators
- Whether any IAA was computed for tasks other than SA
- Adjudication procedure when SA annotators initially disagreed

Requires Expert Verification

- Disagreement rate between professional Luxembourgish editors and benchmark labels on a held-out sample, especially for LA Ortho and HA

Remediation

Gap 1: Annotators are student assistants [Q114] with no professional journalism background; SA Cohen's Kappa 0.45 [Q27] is below typical thresholds; RTE and JUDGEWEL labels are AI-assisted [Q37, Q63, Q64]

Recommendation: Re-annotate held-out HA and LA test sets with at least three professional Luxembourgish editors applying explicit journalistic-register guidelines; report IAA and adjudication procedure